

AMMARA NOOR

AI/ML Engineer | Computer Vision Engineer | Machine Learning Engineer

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Islamabad, Pakistan | Open to remote & onsite | Available immediately

SUMMARY

AI/ML Engineer and Computer Vision Engineer who takes machine learning systems from data collection to production deployment. Built and commissioned two computer vision systems on live textile production lines - an AI garment size measurement station and a YOLOv8 + PatchCore fabric defect detector at 95% accuracy. Shipped production RAG pipelines into client applications and reproduced PatchCore from the paper to 99.56% AUROC on MVTec AD. Hands-on across deep learning, MLOps, model deployment and on-site integration.

TECHNICAL SKILLS

Computer Vision: Object Detection (YOLOv8), Anomaly Detection (PatchCore), Image Segmentation, Image Classification, Object Tracking, Re-ID, Pose Estimation, OCR, Video Analytics, OpenCV

Machine Learning: Deep Learning, PyTorch, TensorFlow, scikit-learn, Hugging Face, Ultralytics, Transfer Learning, Fine-tuning, Model Evaluation (AUROC, mAP, F1, Precision/Recall)

LLM & GenAI: RAG, Embeddings, Vector Search, Chunking & Retrieval Strategy, Prompt Engineering, LLM API Integration, Document AI

MLOps & Deployment: FastAPI, Flask, Docker, MLflow, ONNX, REST APIs, Linux, Git, Latency Profiling, Edge & On-site Deployment

Languages & Web: Python, SQL, JavaScript, C/C++, React.js, Next.js, Node.js, Express.js, MySQL, SQLite, Electron

EXPERIENCE

AI / Computer Vision Engineer

Feb 2026 - Jun 2026

Robionix Technologies

Lahore, Pakistan

- Deployed two production computer vision systems on textile lines, owning each from data collection and model training to on-site installation and operator rollout.
- Collected and labelled a ~1,600-image dataset on the mill floor; tuned detection thresholds under real production conditions to cut false positives.
- Exhibited at the 32nd Textile Asia Expo, leading to evaluation requests from Nishat Mills, Gul Ahmed and Sapphire.

Full-Stack AI Developer

Aug 2025 - Jan 2026

Evolvian Softwares

Lahore, Pakistan

- Built production RAG pipelines end to end - document ingestion, embeddings, vector retrieval, LLM generation - served via REST APIs to React dashboards with role-based auth.
- Rebuilt chunking and retrieval strategy after diagnosing failures standard metrics missed; automated a manual review by chaining LLM, OCR and YOLOv8.

AI & UAV Engineering Intern

Jul 2025 - Sep 2025

National Development Complex (NESCOM)

Islamabad, Pakistan

- Built UAV surveillance with person tracking and vehicle detection on live aerial video, plus a parking dashboard for live slot occupancy.

PROJECTS

MagicQC - AI Garment Size Measurement | PyTorch, OpenCV, Electron, React, Node.js, MySQL

2026

- Camera station that segments each garment, derives 7 points of measure and grades them against buyer tolerance (PASS/FAIL); desktop app plus web app for brands, operators and purchase orders; deployed at MEB Karachi.

Fabric Defect Detection | YOLOv8, PatchCore, PyTorch, Flask, React, Docker

2026

- Fused YOLOv8 (3 defect classes) with PatchCore anomaly detection for unseen defects; 95% accuracy on a held-out set, running live on the mill floor.

Industrial Anomaly Detection - MLOps Pipeline | PyTorch, PatchCore, MLflow, FastAPI, ONNX

2026

- PatchCore reproduced from the paper: 99.56% AUROC on MVTec AD, 87/89 defects caught, F1 0.9775; MLflow-tracked, FastAPI-served, 535 ms CPU latency profiled against a 33 ms target.

RAG Document Pipelines | LLMs, Embeddings, Vector Search, OCR, FastAPI

2025

- Grounded question answering over client documents with cited retrieval, plus an OCR branch extracting structured fields from scanned files.

UAV Surveillance & Smart Parking | YOLO, Object Tracking, Re-ID, OpenCV

2025

- Crowd counting with re-identification, interaction detection and a searchable parking operations dashboard (person, vehicle, face, plate).

RallyLens - Sports Analytics | YOLO, Pose Estimation, Object Tracking

2026

- Player and ball tracking on handheld footage with ball speed, shot counts and an analytics dashboard.

EDUCATION

National University of Technology (NUTECH)

Islamabad, Pakistan

BS Computer Engineering | CGPA 3.44 / 4.00

2022 - 2026

Coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Databases, OOP

ACHIEVEMENTS & LEADERSHIP

- 1st Place, ICAT National Robotics Competition 2025 | Runner-Up, NAIS National AI Seminar 2025
- Chair, IEEE Women in Engineering (NUTECH) | Chief Coordinator, NUTECH Student Council | Millennium Fellow (UN Academic Impact) | Ambassador, 6th International Student Convention 2026